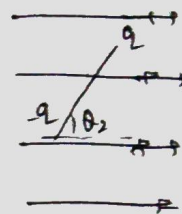
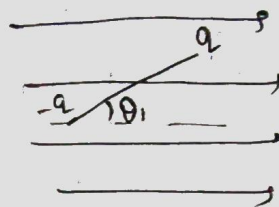


# Potential energy associated with Electric dipole -

$$\boxed{dw = \tau \times d\theta}$$



$$\boxed{\tau = PE \sin \theta}$$

Torque Experienced by Dipole

Energy Associated with Dipole

$$W \int_0^{\theta_2} dw = \int_{\theta_1}^{\theta_2} \tau d\theta$$

$$W = \int_{\theta_1}^{\theta_2} PE \sin \theta d\theta = -PE [\cos \theta]_{\theta_1}^{\theta_2}$$

$$W = -PE [\cos \theta_2 - \cos \theta_1]$$

$$\propto U = -PE [\cos \theta - 1] \quad \text{if } \theta_1 = 0$$

$$\text{if } \theta_1 = 90^\circ, \theta_2 = 0$$

$$\boxed{U = -PE \cos \theta}$$

$$\Rightarrow \boxed{U = -\vec{P} \cdot \vec{E}}$$

$$\theta = 0^\circ \quad \boxed{U = -PE} \text{ (Stable)}$$

$$\theta = 180^\circ \quad \boxed{U = PE} \text{ (Unstable)}$$